Predicting Presidential Approval: An Empirical Analysis of Kalshi Event Contract Accuracy Against RealClearPolitics Polling Averages

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Abstract. Prediction markets centered around political events offer a viable alternative for tracking public opinion, allowing minute-by-minute updates compared to traditional political approval polling. However, the volatility of these markets brings both their interpretability and applicability into question. In order to comprehensively understand the predictive accuracy of these markets, we examine two Kalshi event contracts spanning from September 15, 2025, to January 9, 2026, on U.S. President Donald Trump's approval rating, benchmarking Kalshi's implied approval ratings against the RealClearPolitics polling average (an aggregate of multiple major polls). The results demonstrate that the market-implied estimates closely track realized approval, with the January 1 and January 9 contracts achieving Brier scores of 0.0098 and 0.0457 and Ranked Probability Scores of 0.002 and 0.006, respectively, indicating that the market concentrated almost entirely on the correct outcome. We additionally analyze price movements across these contracts to identify critical inflection points, constructing a narrative for how prediction markets absorb and synthesize political information in real time. Our findings suggest that Kalshi's event contract microstructure produces a more accurate consensus signal, opening further research into the predictive ability and sentiment analysis of prediction markets.

Keywords: prediction markets · public opinion · presidential approval · Kalshi · polling

1 Introduction

The measurement of public opinion has long been the domain of professional polling sites, which aggregate survey responses into approval ratings for political leaders. In particular, the RealClearPolitics (RCP) average—a rolling unweighted aggregate of multiple major polls—serves as the widely cited consensus benchmark for such organizations. However, these polls carry several structural limitations: survey responses are collected over days or weeks and published with inherent temporal lag, and the aggregation of polls with differing methodologies can introduce smoothing that obscures potentially meaningful short-term fluctuations. As a result, A figure published today may reflect sentiment which diverges

from true public opinion, and sudden political events are usually absorbed only gradually into the aggregate.

Prediction markets offer a fundamentally different mechanism for information aggregation. Unlike opinion polls, which capture self-reported preferences with no implied consequence for misreporting, prediction market participants face direct monetary penalties for miscalibrated positions. Moreover, traditional polling services suffer from implied biases from participants, organizations, and even government administrations which may sway results to a degree. All prediction markets operate under the same playing field, eradicating such concerns. Kalshi, being the first federally regulated prediction market exchange in the United States, is placed in a unique position to utilize these principles in order to effectively aggregate information more accurately due to wider usage. In particular, U.S. Presidential approval markets allow traders to continuously express probabilistic beliefs at minute-level resolution, in contrast to the typically slower publication cadence of traditional polls.

In order to further establish whether these markets provide a more favorable approach to establishing approval consensus, we analyze 1-minute-level pricing data for two U.S. presidential approval event contracts ranging the last 4 months and construct implied ratings from normalized bucket prices at each timestamp. We benchmark these market-implied estimates against the realized RCP average on each settlement date, evaluating accuracy with proper scoring rules (Brier Score and Ranked Probability Score) in order to assess convergence dynamics over each contract's lifetime. We additionally apply an adaptive jump detection algorithm to identify significant market repricing events and test if Kalshi markets can serve as leading indicators of political sentiment. Our results demonstrate that both contracts achieved near-zero proper scoring rule values, substantially outperforming random guessing baselines, and that significant market repricing events frequently led the RCP polling average by days, showcasing Kalshi's capability to act as a leading provider on public consensus towards world events.

2 Methodology

2.1 Data

RealClearPolitics Polling Average: The unweighted average of daily approval and disapproval percentages for President Trump for a select number of major polls. The RCP average serves as our consensus benchmark for the realized approval rating.

Kalshi Market Data: 1-minute-level pricing data for two categorical event contracts on Trump's approval rating from Kalshi:

- "What will Trump's approval rating be on January 1, 2026?" [1]
- "What will Trump's approval rating be on January 9, 2026?" [2]

The January 1 contract was active from September 15, 2025, providing 108 days of trading data, while the January 9 contract was active from January 3, 2026, providing 7 days. The January 1 contract was split into 10 internal buckets, each 0.9 percentage points wide from 40% to 50% (e.g., "42.0 to 42.9"). The January 9 contract covers 6 internal buckets from 42% to 44.3%, each 0.3 percentage points wide (e.g., "42.8 to 43.1"). Each contract contained 2 additional edge buckets (e.g., "41 or below", "50 or above") added to capture the tails of the distribution. At every timestamp, each bucket additionally had its own price, which reflected the market's belief that Trump's RCP approval would land in that range.

2.2 Probability Estimation Framework

At each timestamp, contract prices across all buckets were normalized to sum to one. From this normalized price distribution, we computed a market-implied approval point estimate:

$$\hat{A}_t = \sum_{i=1}^{n} p_{i,t} \cdot m_i \tag{1}$$

where $\hat{A}_t$ is defined as the market-implied approval rating at time t, $p_{i,t}$ is the normalized probability of bucket i at time t, and m_i is the midpoint of bucket i. Midpoints were calculated as the average of the lower and upper bounds for each bucket, with edge buckets being extended by one bucket width to each bound in the open direction.

2.3 Forecast Scoring

To evaluate the predictive accuracy of these markets, each contract's final forecast distribution was scored against the actual RCP rating. To mitigate noise, we defined the final distribution as the mean of the normalized probability vectors over the final hour of trading. The RCP approval percentage on the target date was then mapped to the corresponding bucket to produce a one-hot vector o, where the correct bucket k sets $o_k = 1$ and all others $o_j = 0$.

To quantitatively showcase market predictability, we utilized the **Brier Score**[3], which measures the mean squared error between the predicted probability vector and the observation vector:

$$BS = \frac{1}{2} \sum_{i=1}^{n} (p_i - o_i)^2 \tag{2}$$

where p_i is the forecast probability for bucket i when the market resolves, and o_i is the aforementioned vector. Our results are halved to normalize the metric's range from 0 (perfect accuracy) to 1 (perfect inaccuracy).

We additionally utilized the **Ranked Probability Score (RPS)**, which penalizes forecasts proportionally to their cumulative distance from the realized outcome:

$$RPS = \frac{1}{n-1} \sum_{i=1}^{n} (F_i - O_i)^2 \tag{3}$$

where $F_i = \sum_{j=1}^{i} p_j$ and $O_i = \sum_{j=1}^{i} o_j$ are the cumulative sums of the predicted probabilities and observed outcomes, respectively. Unlike the Brier Score, RPS also accounts for how far the predicted outcomes are from the correct bucket. The score ranges from 0 to 1, with 0 equating to a perfect forecast and 1 representing the worst possible forecast.

2.4 Jump Detection

In order to identify significant movements in the market-implied approval time series $\{\hat{A}_t\}$, we developed a jump detection algorithm which focused on the aforementioned data density metrics to inform the scaling of detection parameters, including ticks per day, median gap between ticks, and total time span in days. Tick-over-tick changes $\Delta_t = \hat{A}_t - \hat{A}_{t-1}$ were utilized to construct a rolling z-score:

$$z_t = (\Delta_t - \mu_w)/\sigma_w \tag{4}$$

where μ_w and σ_w are the rolling mean and standard deviation, respectively, over a window of size w sized at 5% of total ticks, clipped between 20 and 100, with a minimum period requirement of one-fifth the window size. Scaling the window by 5% adapts the window to each market's trading volume, so the more heavily traded January 1st market receives larger windows while the thinly traded January 9th market receives smaller windows. A floor of 20 ticks also ensures there exist enough observations to construct a robust z-score, while a cap of 100 ticks prevents the window from growing so large that it absorbs meaningful changes. The minimum period of $w/5$ additionally prevents z-scores from being computed on too few observations.

Price jumps were identified using two distinct methods: rolling z-scores and absolute change thresholds. Both techniques are adapted to Kalshi contract trading frequency and duration, and detect different aspects of price jump. In particular, utilizing an absolute threshold helps catch genuinely large moves that might have a low z-score becaue of market noise. With this in mind, our z-score threshold is conservatively placed at 4.5, and is relaxed for contracts with denser trading activity (more reliable statistical estimates). Moves were also required to exceed a minimum size relative to the contract's duration to filter out noise in short-lived markets. Nearby detections within a one to two-hour window were consolidated to avoid multiple counts of single movements. Jumps were then classified as "major" if they exceeded the median detected jump size for that contract or

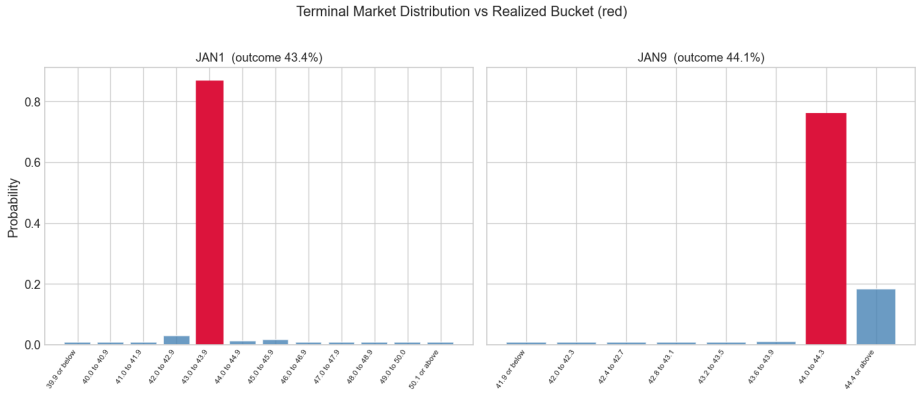

Fig. 1. Terminal market probability distributions for the two contracts. The red bar in each panel marks the realized bucket; blue bars show probability assigned to non-realized buckets. JAN1 outcome: 43.4%; JAN9 outcome: 44.1%.

represented the largest single directional move, and "minor" otherwise. Internal sensitivity testing was conducted across a range of threshold values to verify that results were robust to parameter choice.

3 Results

3.1 Forecast Accuracy

In both the January 1st and January 9th markets, Kalshi's algorithmically implied approval ratings showcased strong predictive accuracy in relation to the RCP outcome, as shown in Fig 1. The January 9th market displayed the stronger performance out of the two markets that we used for our research, yielding a Brier score of .0096 and an RPS of .006, with 77% of its probability mass being concentrated on the correct result bucket (44.0–45.9%) against the actual result of 44.1%. The remaining 23% exhibited a spread: 18% on 46.0–47.9, 3% on 48.0–49.9% and the remaining 2% distributed across all other buckets. The January 1st market yielded a Brier score of .1926 and an RPS of .002, with the market placing 90% of its probability mass in the bucket 44.0–45.9% against the actual result 43.4%. Only 4% of the probability mass was allocated into the 42.0–43.9% bucket.

While this does represent weaker performance, both markets still outperformed the random guessing baseline by 78.3%. The higher Brier score reflects a bucket "boundary effect" rather than a miscalibration—the market's modal estimate overshot the actual outcome by .6 percentage points—landing just outside of the correct bucket. Additionally, the market's RPS of 0.002 indicates that event's probability mass was clustered near the correct outcome rather than dispersed

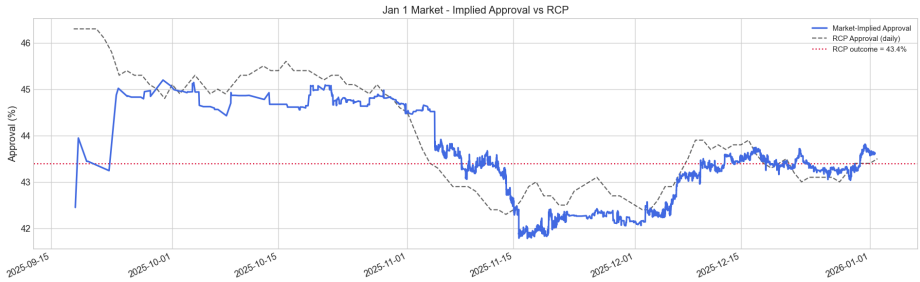

Fig. 2. Jan 1 Market: Market-implied approval (blue) vs. RCP polling average (dashed) vs. realized outcome (red dotted line at 43.4%). The market tracks the general trend and converges to within a fraction of a percentage point of the outcome across 108 days of trading.

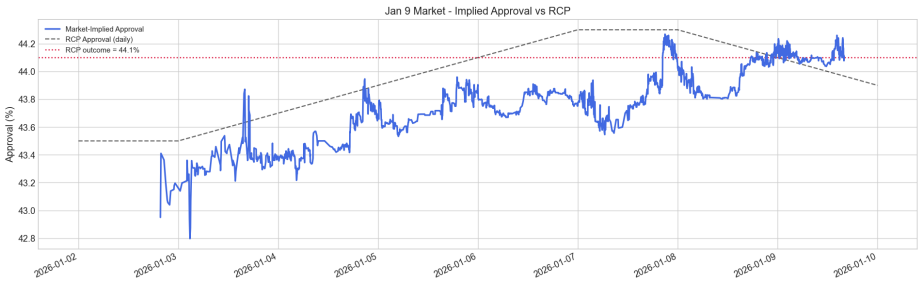

Fig. 3. Jan 9 Market: Market-implied approval (blue) vs. RCP polling average (dashed) vs. realized outcome (red dotted line at 44.1%). Over 7 days of trading the market converges sharply to the realized outcome.

across the outcome space. Taken together, the two contracts demonstrate that Kalshi's event market microstructure produces a well-calibrated forecast signal across both a long-horizon contract (108 days) and a short-horizon contract (7 days), as seen in Fig 2 and 3.

3.2 Market Convergence

Both Kalshi event contracts exhibit a consistent pattern of error reduction as the resolution date approaches, consistent with markets efficiently incorporating information over time.

The January 1st market displayed the longest convergence path of the two markets, with a peak absolute error of 1.75–1.80 percentage points(seen in early October 2025). Error declined at a steady rate through October and November before a noticeable secondary spike that took place in mid-November. Kalshi repriced downward immediately, converging towards the actual 43.4% outcome. From December onwards, Kalshi's absolute error remained below .5 percentage

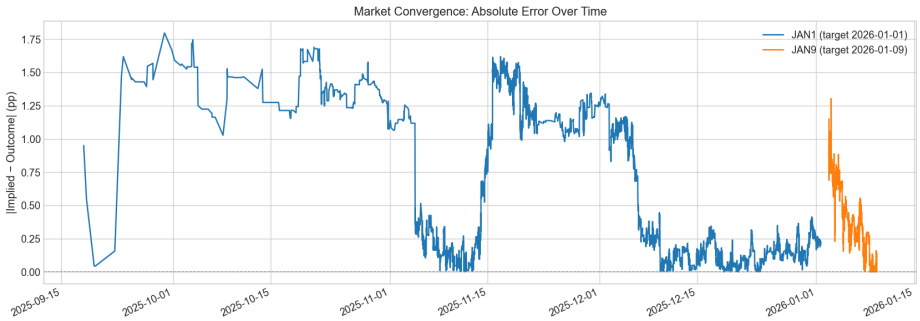

Fig. 4. Market convergence: absolute error $|\hat{A}_t - \text{outcome}|$ over time for the JAN1 (blue) and JAN9 (orange) contracts. Both long- and short-horizon contracts display systematic error reduction toward zero as resolution approaches.

points and approached 0 at the point of resolution. The January 9th market, despite a 101 day difference in its trading window, followed a similarly disciplined convergence path, peaking at approximately 1.30 percentage points in early January before converging sharply towards 0 by the resolution on January 9th. Kalshi's ability to hastily reprice from absolute error to near-zero percentage points in under a week indicates that the platform's markets can rapidly incorporate available information, even under extreme time constraints.

In contrast to Kalshi's predictive calibration, the RCP polling average does not exhibit the same steady convergence behavior. While it does travel to the actual outcome, the polling averages oscillate without a consistent pull toward the resolution—an obvious sign of the damaging effect of the lag inherent in survey-based analysis.

3.3 Inflection Point Analysis

The adaptive jump algorithms identified various statistically significant price movements in the market implied approval timeframe. Here, we examine only the median threshold algorithm, given the two provide similar results. The JAN1 contract produced 16 major jumps out of 31 detected, while the JAN9 contract produced 17 major jumps out of 33 detected—a notably high density of major movements given JAN9's 7-day window, reflecting heightened market activity around the inauguration period.

The most notable inflection point in the January 1st market occurred in mid-November 2025. This took place due to an implied approval drop of approximately .7 percentage points—the single largest movement across both markets. This repricing displayed a corresponding decline in the RCP polling average by several days, providing direct evidence that the market was impacted by negative political information before it was reflected in the RCP survey results. Again, in

December we witnessed another moment in which the market reacted to political information before the RCP survey results were able to display it.

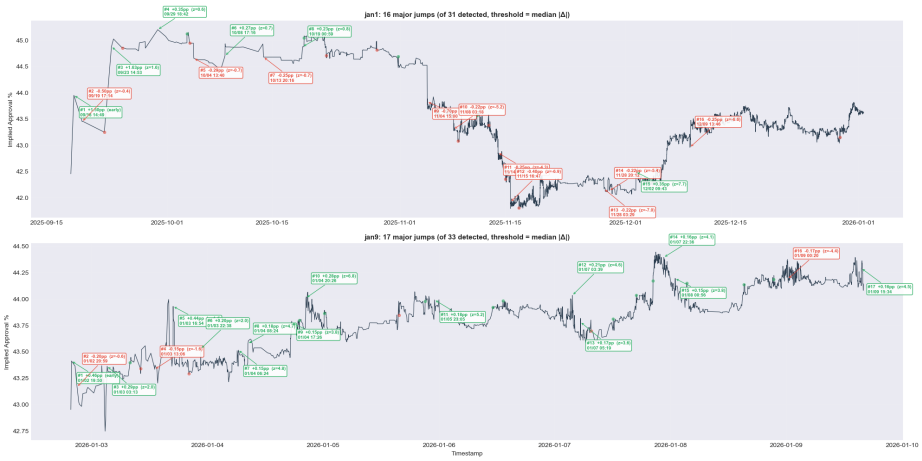

Fig. 5. Major jump classification across all contracts (Median Threshold Method). Jumps are classified as "major" (labeled, above median detected size) or "minor" (small dot only). Green annotations indicate upward repricing events; red indicate downward events. Each label reports the sequential jump number, magnitude, z-score, and timestamp.

In the January 9th market, the most notable activity occurred on January 3–4, 2026. A sharp drop of −0.46 percentage points on January 3 was followed by a spike of +0.44 percentage points on January 4. This is consistent with markets processing and then recalibrating due to inauguration-adjacent news. In addition to the spike—downward and then upward, respectively—on January 3–4, we can also see a cluster of percentage point changes in the form of upward jumps on January 8th—+0.16 pp, +0.13 pp, +0.12 pp—which evidently drove the market implied approval toward its terminal value near the actual outcome of 44.1%.

In both markets, the inflection point analysis provides quality evidence of the granularity advantage. These Kalshi markets moved in the correct direction ahead of the RCP polling average, which is clear evidence of a data analysis advantage as Kalshi's algorithm processed the information minute-by-minute rather than the multi-day lag inherent in the survey aggregation that the RCP provides the general public.

4 Discussion

4.1 Kalshi Outperforms Expert Aggregation

The results presented in the previous section suggest that Kalshi's contract microstructure produces a more accurate and more timely political consensus signal than traditional expert aggregation. Two distinct advantages emerge from the data. The first is **accuracy**: the Brier scores of 0.0096 and 0.1926 across the January 9th and 1st markets both dramatically outperformed the random guessing baseline of 0.25. This demonstrates that financially incentivized market participants arrive at a better estimate of actual approval than the RCP polling aggregate alone. The second is **granularity**: where the RCP polling average updates at best daily and reflects survey data that was obtained days prior to its release, Kalshi markets update minute-by-minute, repricing continuously as new political information enters the market.

The mechanism underlying both advantages is the financial incentive structure of prediction markets. Unlike expert forecasters or polling aggregators, Kalshi participants are directly penalized for being wrong (i.e. a miscalibrated position costs the user money). This skin-in-the-game dynamic disciplines market participants in a manner that no survey is able to replicate. The inflection point analysis makes this mechanism very apparent; when the market detected a shift in political sentiment in mid-November 2025, it repriced by 0.70 percentage points in a single movement before the RCP average had moved at all. The polling average caught up days later—but the market had already processed the information.

4.2 January 1st Interpretation

The January 1st market should be discussed specifically given its higher Brier score of 0.1926 relative to January 9th's Brier score of 0.0096. The market placed 90% of its terminal probability on the 44.0–45.9% bucket, while the realized outcome of 43.4% fell just across the boundary in the 42.0–43.9% bucket. This should not be interpreted as a failure of the market mechanism. Two points support this reading. First, the RPS of 0.002—lower than January 9th's 0.006—confirms that the market's probability mass was tightly clustered near the correct outcome; the miss was a matter of a 0.6 percentage point overshoot at a bucket boundary, not a broadly miscalibrated distribution. Second, the market still outperformed the random baseline by 78.3%, meaning it was far more informative than chance even in its weakest showing.

The January 1st result displays a clear structural limitation for discrete bucket markets: when the true result falls near a bucket boundary, even a well-calibrated market will be penalized disproportionately by a Brier score test. In order toimprove measured accuracy without requiring any change to the underlying forecasting mechanism, we would need to tighten intervals around the likely result range.

4.3 Methodology Limitations

This study has several limitations worth acknowledging. The observation window covers a single political figure, constraining the generalizability of the findings. Additionally, benchmarking against a polling aggregate rather than individual polls equates to comparing our markets against a smoothed signal that itself introduces its own lag. Finally, liquidity constraints in smaller Kalshi markets may affect price efficiency, particularly in the early days of a market's life when trading volume is thin. We can visualize this specifically in the elevated early-period errors in both the January 1st and January 9th convergence charts.

5 Conclusion

This paper examined whether Kalshi's categorical event contracts on U.S. Presidential approval ratings produce accurate and timely political forecasts relative to the RealClearPolitics polling average. By analyzing the January 1 and January 9 categorical Trump approval contracts, our results demonstrate that prediction market prices generate well-calibrated probability distributions that converge towards realized outcomes. The contracts both considerably outperformed random guessing, with Brier and RPS scores indicating probability mass concentration near the correct approval rating. Further jump detection also showed that market prices led RCP average movements, with the largest 0.70 pp jump in mid-November 2025 occurring days before the respective RCP sentiment shift. The January 9 market also achieved sharp convergence in only seven days, despite opening at 1.3 pp away from the eventual settlement. These outcomes align with the Hayekian hypothesis, where Kalshi's financially incentivized market structure aggregates dispersed information at a greater accuracy and speed than traditional survey-based aggregation. Ultimately, this research has implications for how political analysts, campaign strategists, journalists, and policymakers might benefit from supplementing polling data with prediction markets. While we present comprehensive results on categorical Kalshi approval contracts, there are also many directions for future research. First, extending the contracts to a broader set of political figures, approval rating contexts, and jurisdictions would help generalize our findings beyond the specific Trump timeframe across different liquidity and volatility levels. Second, incorporating contracts with finer bucket intervals could mitigate the boundary effects that complicate our Brier Score interpretation. Lastly, comparative analysis over other prediction markets (e.g., Polymarket, PredictIt) could help guide designers and regulators in determining what market features optimize forecast accuracy.

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